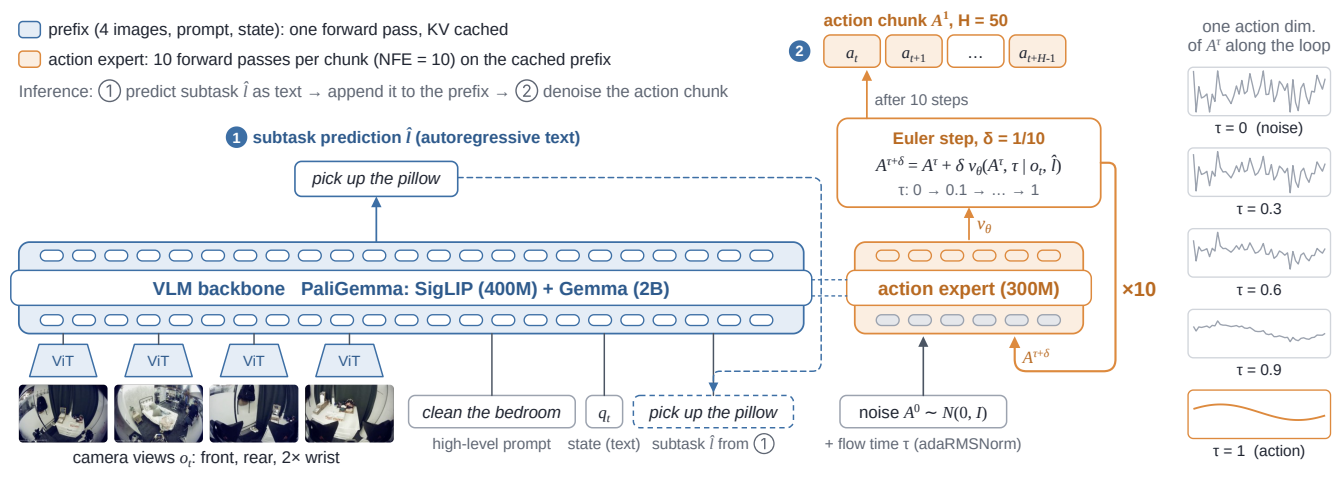

prefix (4 images, prompt, state): one forward pass, KV cached
action expert: 10 forward passes per chunk (NFE = 10) on the cached prefix
Inference: ① predict subtask $\hat{l}$ as text → append it to the prefix → ② denoise the action chunk
action chunk A^1, H = 50
2
a_t
a_{t+1}
…
a_{t+H-1}
one action dim. of A^τ along the loop
after 10 steps
Euler step, δ = 1/10
$A^{\tau+\delta} = A^\tau + \delta\, v_\theta(A^\tau, \tau \mid o_t, \hat{l})$
τ: 0 → 0.1 → … → 1
v_θ
1 subtask prediction $\hat{l}$ (autoregressive text)
pick up the pillow
VLM backbone PaliGemma: SigLIP (400M) + Gemma (2B)
action expert (300M)
×10
$A^{\tau+\delta}$
ViT
ViT
ViT
ViT
clean the bedroom
q_t
pick up the pillow
noise $A^0 \sim N(0, I)$
camera views o_t: front, rear, 2× wrist
high-level prompt
state (text)
subtask $\hat{l}$ from ①
+ flow time τ (adaRMSNorm)
τ = 0 (noise)
τ = 0.3
τ = 0.6
τ = 0.9
τ = 1 (action)